

HAR Official X-Fi Comparison

Blue indicates higher accuracy than official X-Fi; red indicates lower accuracy.

Method	Split	Source	Modality	Ours Acc.	X-Fi Acc.	Δ	Macro-F1
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk	3.40	26.50		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth	83.71	48.10		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	lidar	3.36	52.70		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	mmwave	66.72	85.70		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth	83.50	45.30		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+lidar	3.91	35.20		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+mmwave	66.46	73.40		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+lidar	83.37	51.60		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+mmwave	95.76	79.80		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	lidar+mmwave	66.69	88.70		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+lidar	83.66	48.70		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+mmwave	95.66	70.70		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+lidar+mmwave	66.39	77.80		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	depth+lidar+mmwave	95.64	80.50		
VK-RCD-HAR-Teacher	random_split	teacher_random_all_combinations.csv	vk+depth+lidar+mmwave	95.57	72.20		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk	65.31	26.50		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth	88.02	48.10		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	lidar	40.80	52.70		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	mmwave	85.88	85.70		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth	89.51	45.30		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+lidar	68.54	35.20		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+mmwave	92.75	73.40		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+lidar	88.52	51.60		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+mmwave	96.45	79.80		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	lidar+mmwave	89.51	88.70		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+lidar	89.62	48.70		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+mmwave	96.56	70.70		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+lidar+mmwave	92.73	77.80		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	depth+lidar+mmwave	96.44	80.50		
VK-RCD-HAR-Student-VK-Missing	random_split	student_vk_random_all_combinations.csv	vk+depth+lidar+mmwave	96.56	72.20		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk	3.40	26.50		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth	83.71	48.10		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	lidar	3.45	52.70		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	mmwave	66.72	85.70		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth	83.50	45.30		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+lidar	3.85	35.20		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+mmwave	66.46	73.40		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+lidar	83.39	51.60		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+mmwave	95.76	79.80		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	lidar+mmwave	66.64	88.70		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+lidar	83.70	48.70		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+mmwave	95.66	70.70		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+lidar+mmwave	66.33	77.80		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	depth+lidar+mmwave	95.62	80.50		
VK-RCD-HAR-Teacher	random_split	all_combinations.csv	vk+depth+lidar+mmwave	95.59	72.20		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk	60.94	26.50		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth	86.85	48.10		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	lidar	33.14	52.70		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	mmwave	83.13	85.70		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth	88.09	45.30		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+lidar	63.63	35.20		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+mmwave	91.05	73.40		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+lidar	87.24	51.60		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+mmwave	95.57	79.80		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	lidar+mmwave	85.92	88.70		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+lidar	88.62	48.70		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+mmwave	95.96	70.70		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+lidar+mmwave	91.23	77.80		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	depth+lidar+mmwave	95.50	80.50		
VK-RCD-HAR-Ablation-NoKD	random_split	ablation_no_distill_all_combinations.csv	vk+depth+lidar+mmwave	95.99	72.20		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk	5.52	26.50		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth	85.59	48.10		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	lidar	2.65	52.70		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	mmwave	69.91	85.70		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth	86.48	45.30		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+lidar	4.58	35.20		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+mmwave	67.75	73.40		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+lidar	86.16	51.60		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+mmwave	95.54	79.80		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	lidar+mmwave	67.80	88.70		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+lidar	86.43	48.70		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+mmwave	95.50	70.70		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+lidar+mmwave	67.09	77.80		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	depth+lidar+mmwave	95.42	80.50		
VK-RCD-HAR-Ablation-UniformFusion	random_split	ablation_uniform_fusion_all_combinations.csv	vk+depth+lidar+mmwave	95.39	72.20		